

Operation Readiness: Detection & Response

A Strategic Briefing on Course 6
of the Google Cybersecurity
Professional Certificate.

AUTHORIZATION:

Google / Coursera

CLEARANCE:

Beginner Level
(No Prior Experience)

ENROLLMENT_STATUS:

Active (315,789+ Deployed)

RATING:

4.8/5.0

Status Board



The Threat Landscape

Malicious actors are continuously probing networks. Defining a security incident and acting swiftly is the frontline of enterprise defense.



The Mission Objective

Equip analysts to master the incident response lifecycle. Analyze network traffic, interpret IDS/SIEM logs, and execute containment, eradication, and recovery.



The Operational Output

Job-ready capability for entry-level cybersecurity roles, validated by Google industry experts.

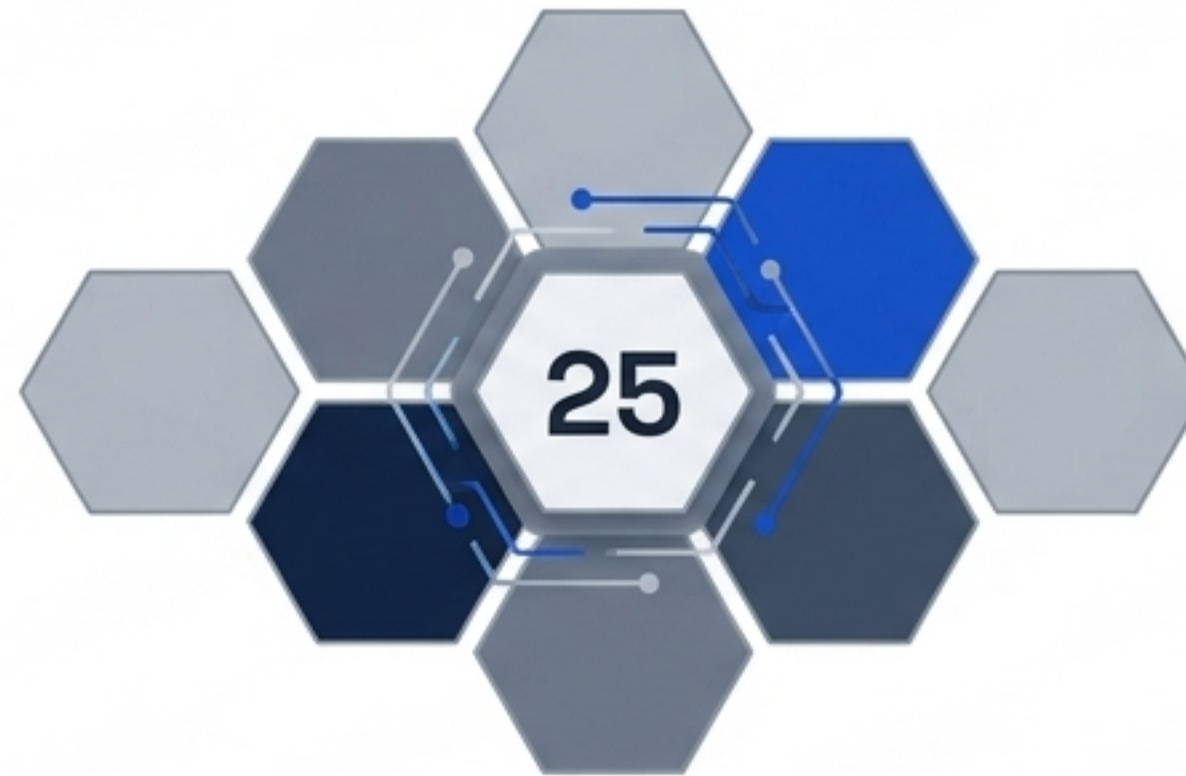
Time-to-Value Logistics

Deployment Schedule

DURATION:
2 Weeks

VELOCITY:
10 Hours / Week
(Self-paced)

STRUCTURE:
4 Tactical Modules
(18 Total Hours)



Hands-on Assignments
& Artifact Investigations

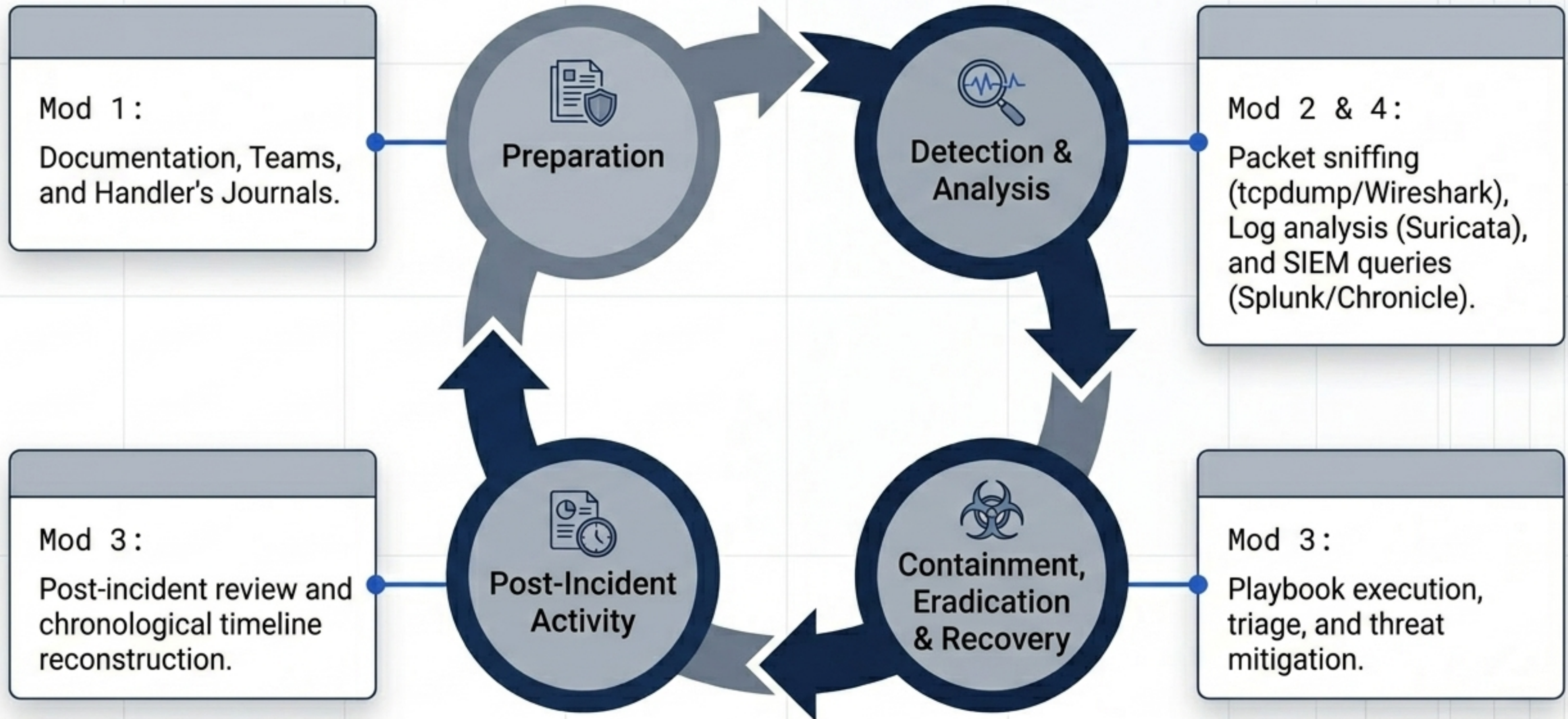


98% Learner
Satisfaction

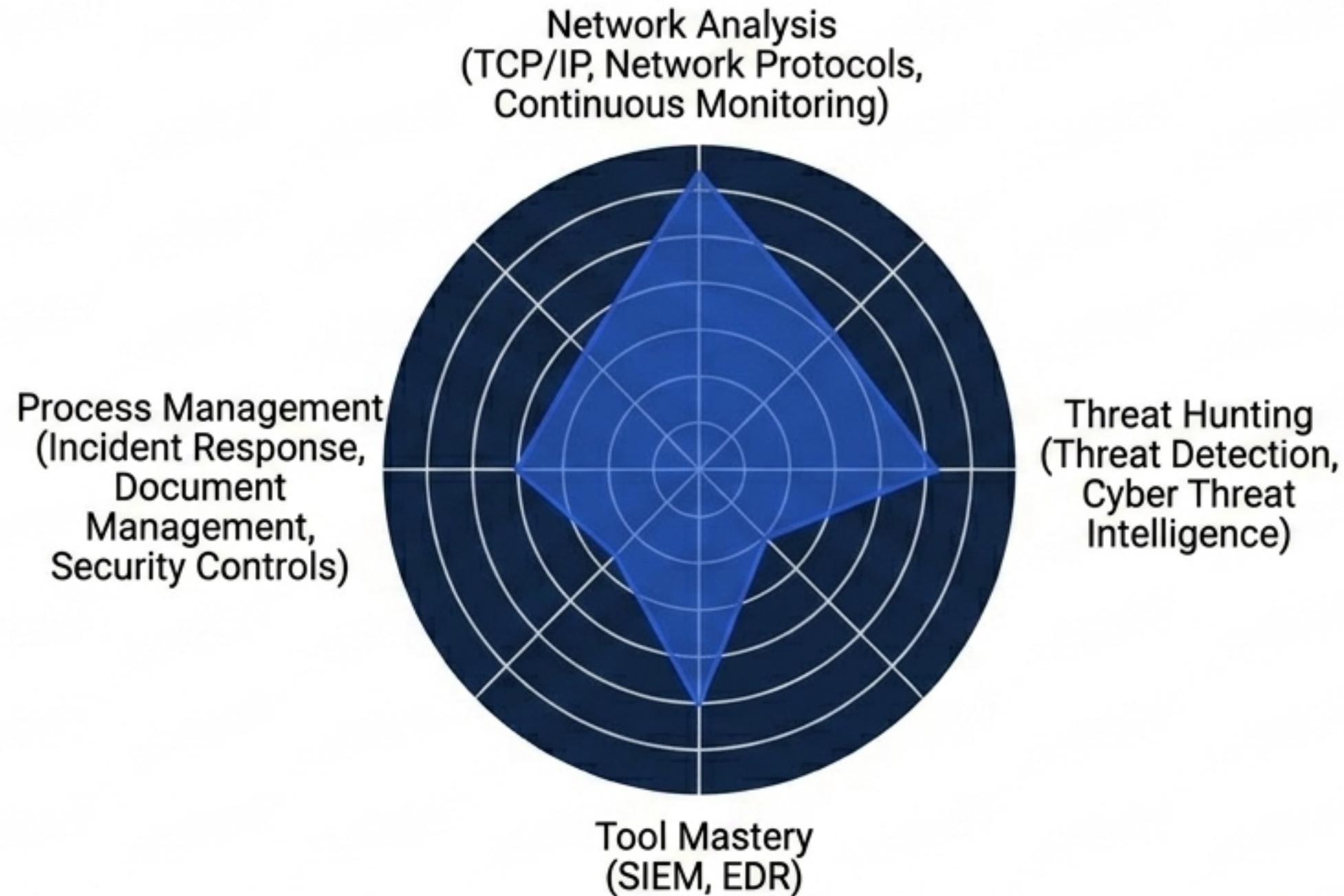


Shareable LinkedIn Credential.
Rapid upskilling available in 12
languages, designed for immediate
application in SOC environments.

The Playbook: NIST Framework Alignment



Target Competency Profile



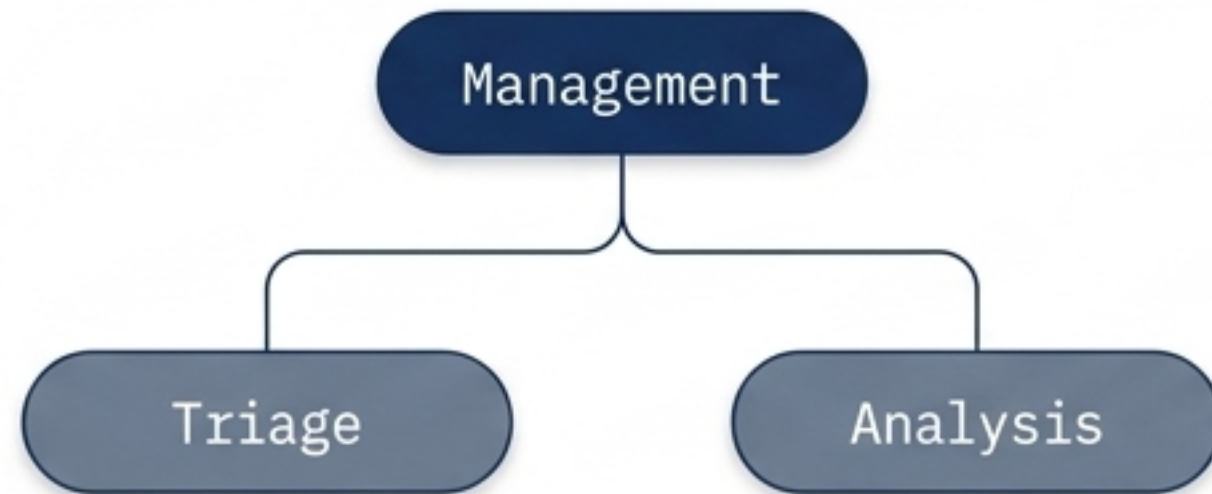
Develops a “T-Shaped Analyst”—broad foundational knowledge of computer security architecture, combined with deep, tactical expertise in network monitoring and event investigation.

The Analyst Arsenal: Tool Matrix

Packet Analyzers (Sniffers)	Intrusion Detection (IDS)	SIEM & SecOps	Threat Intelligence
tcpdump, Wireshark	Suricata	Splunk, Google SecOps (Chronicle), Wazuh	VirusTotal
Capture and filter raw network traffic.	Match network traffic against signature rules.	Aggregate, index, and correlate logs globally.	Analyze suspicious file hashes and URLs.
Inspecting packet headers for malicious communications.	Generating alerts for suspicious network events.	Executing complex search queries to investigate events.	Verifying indicators of compromise (IoCs).

Phase 1: IR Foundations & Triage

The Human Element



Establishing the Incident Response Lifecycle. Defining security incidents and cross-team communication protocols. protocols.

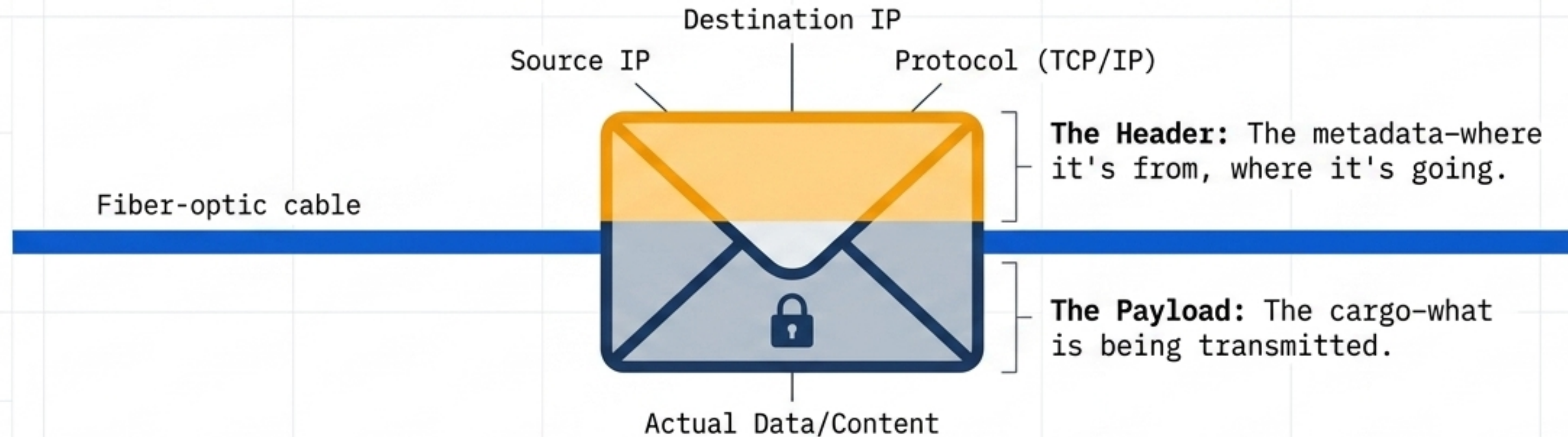
The Documentation



The Incident Handler's Journal. Mastering the critical soft skill of documenting evidence, alerts, and management workflows accurately.

DURATION: 3 Hours | FOCUS: Preparation & Roles

Phase 2: Network Monitoring & Packet Analysis



tcpdump:

Command-line filtering of traffic flows.

Wireshark:

Deep-dive GUI inspection to identify data exfiltration attacks.

DURATION: 5 Hours | FOCUS: Traffic Flow & Sniffing

Phase 3: Investigation & Artifact Analysis

DETECTION

4D 5A 90 00 03 00 00 00 00 04 00 00 00 FF FF 00 00 B8
00 00 00 00 00 00 00 40 00 00 00 00 00 00 00 00 00



Utilizing VirusTotal to analyze suspicious file hashes and verify Indicators of Compromise (IoCs).

THE TIMELINE



Approximating event chronology and documenting evidence with chain of custody forms.

RESPONSE

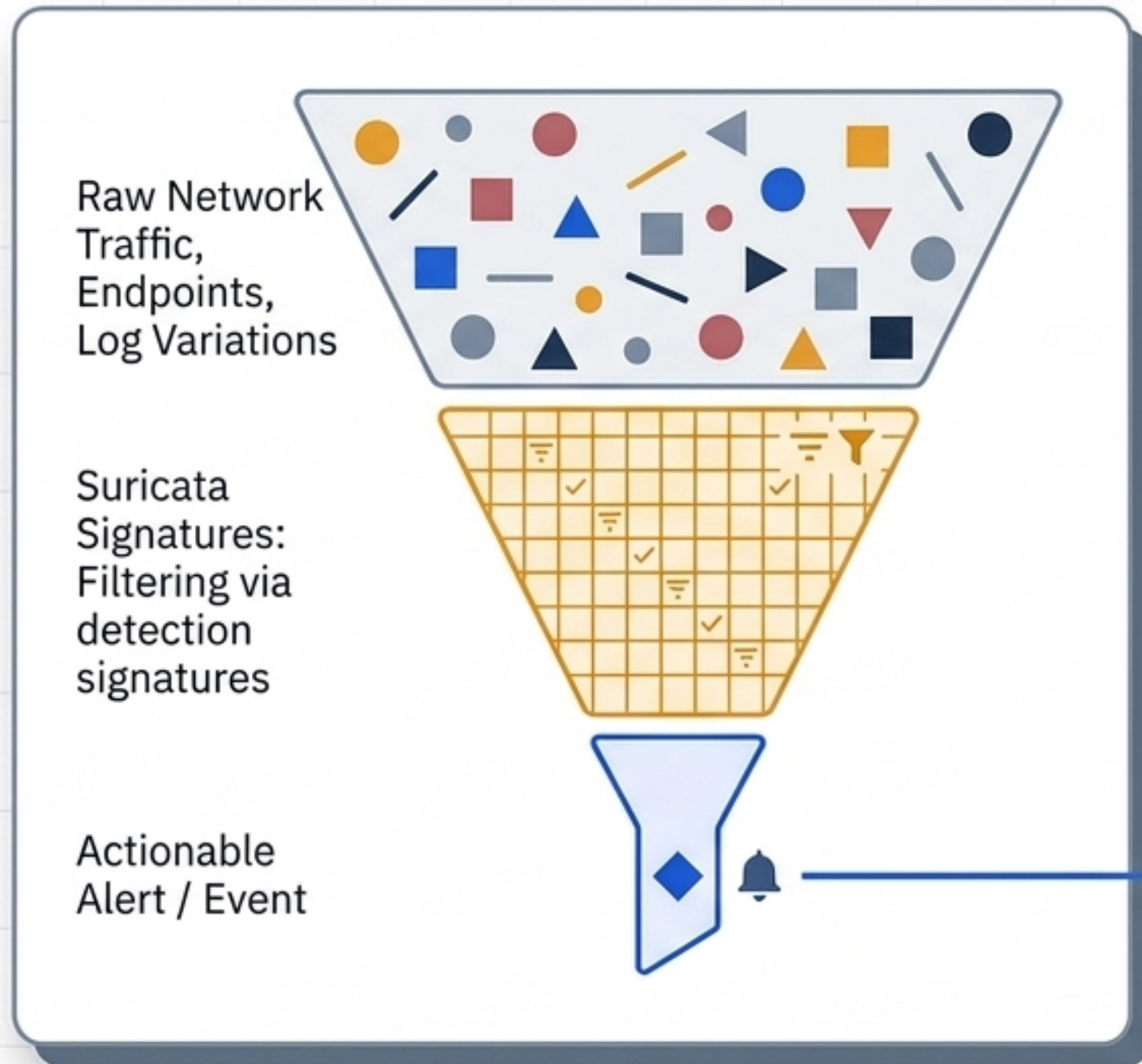


RESPONSE

Executing cybersecurity playbooks for specific threats (e.g., Phishing) to contain, eradicate, and recover operations safely.

DURATION: 4 Hours | FOCUS: Playbooks & Containment

Phase 4: IDS & SIEM Mastery

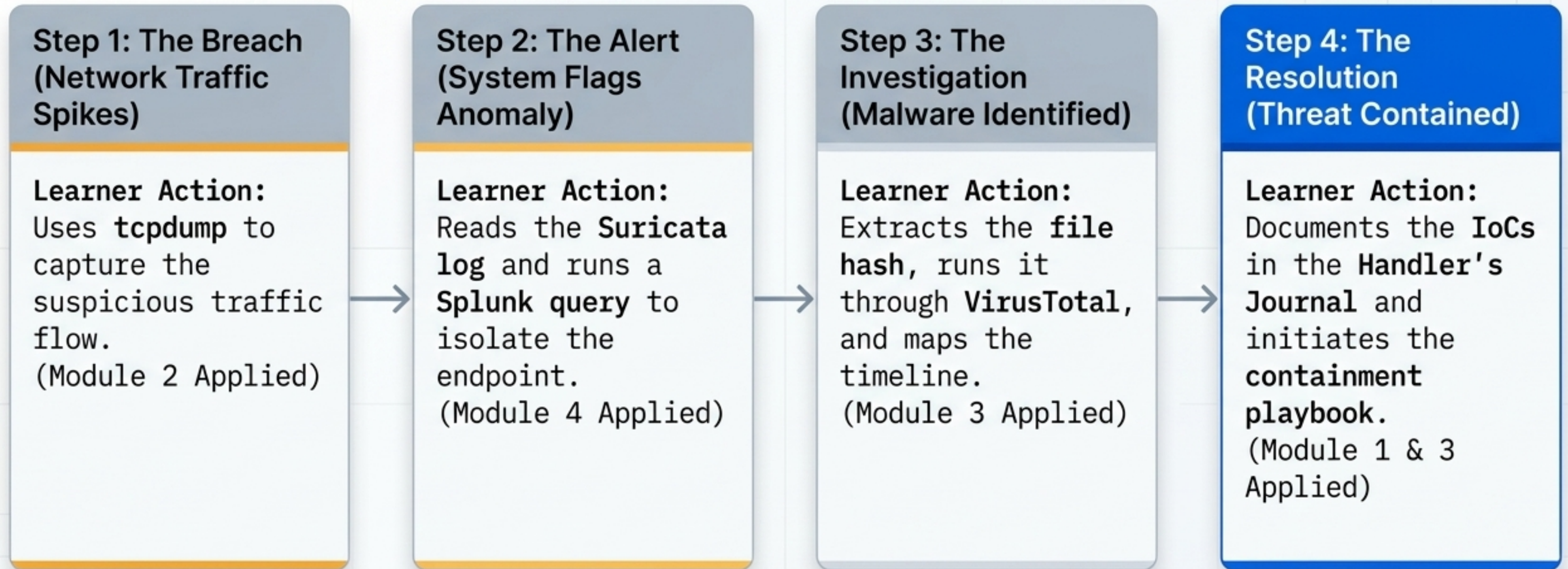


```
> splunk: search index=main  
sourcetype=suricata* alert_level>2  
  
> gcp_secops: select * from events where  
rule_id='MALWARE_DETECTION'  
  
> wazuh: /var/ossec/bin/wazuh-logtest -f  
/var/log/syslog --custom-rules=rules.xml  
  
> correlating events across global  
infrastructure...  
  
> isolating threats: IP 192.168.1.100, User  
'admin', Process 'powershell.exe'
```

Performing advanced search queries in Splunk, Google SecOps (Chronicle), and Wazuh to correlate events and isolate threats globally.

DURATION: 6 Hours | FOCUS: Log Aggregation & Queries

Synthesis: Anatomy of an Investigation



Field Intelligence: Outcomes & Validation

315,789+

Enrolled globally

4.8/5.0

Rating from 3,360 reviews

98%

Learner satisfaction

Learner Comm-Links

"To be able to take courses at my own pace and rhythm has been an amazing experience."

– Felipe M.

"I directly applied the concepts and skills I learned from my courses to an exciting new project at work."

– Jennifer J.

"The professor's teaching makes complex topics easy to understand. Highly recommended!"

– ND

Mission Debrief: Deployment Readiness



Industry Validated

Curated and taught by Google industry experts to meet current market demands.

Enterprise Ready

Hands-on portfolio artifacts (e.g., Incident Handler's Journal) prove practical capability to employers.

Career Accelerated

Adds a globally recognized, shareable credential to LinkedIn profiles, unlocking entry-level SOC roles.